The Fourier Basis of Digit Arithmetic: Mechanistic Interpretability of Addition Circuits in Language Models

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Abstract

How do language models perform integer addition? We present a mechanistic analysis of the ones-digit addition circuit across three architectures (Gemma 2B, Phi-3 Mini, LLaMA 3.2-3B), revealing that digit identity is encoded in a **9-dimensional Fourier subspace** corresponding to the irreducible representations of $\mathbb{Z}/10\mathbb{Z}$. We establish a complete causal evidence triangle: the Fourier subspace is *sufficient* (Fisher activation patching transfers 100% of digit information), *necessary* (multi-layer ablation reduces accuracy to near-chance), and *specific* (matched-dimension random controls have zero effect). We discover that models *progressively rotate* this subspace across layers to align with the unembedding matrix for readout, and that different architectures emphasize different components of the Chinese Remainder Theorem decomposition—Gemma emphasizes mod-2 (parity), Phi-3 emphasizes mod-5, and LLaMA uses a balanced ordinal encoding. Finally, we show that Fourier phase rotation within this subspace can *steer* digit predictions with 70–88% exact shift accuracy when combined with unembedding-informed corrections, demonstrating genuine mechanistic control over arithmetic computation.

1 Introduction

Language models can perform integer arithmetic [Brown et al., 2020, Touvron et al., 2023], yet the internal computational mechanisms remain poorly understood. Prior work on mechanistic interpretability has identified circuits for indirect object identification [Wang et al., 2022] and modular arithmetic in trained transformers [Nanda et al., 2023, Zhong et al., 2024], but the arithmetic circuits of *pretrained* language models operating on natural-language-formatted problems have received comparatively little attention.

We present a comprehensive mechanistic analysis of the **ones-digit addition circuit** across three pretrained architectures: Gemma 2B [Team et al., 2024], Phi-3 Mini [Abdin et al., 2024], and LLaMA 3.2-3B [Dubey et al., 2024]. Our central finding is that these models encode digit identity in a **9-dimensional Fourier subspace** of the residual stream that corresponds exactly to the irreducible representations of $\mathbb{Z}/10\mathbb{Z}$—the cyclic group of integers modulo 10.

Contributions. We make five main contributions:

1. **Fourier subspace characterization.** We show that the top-9 SVD directions of per-digit mean activations at computation layers form a perfect discrete Fourier basis of $\mathbb{Z}/10\mathbb{Z}$, with frequency purities of 72–99% across three models (§4.1).

2. **Complete causal evidence.** We establish necessity via multi-layer Fourier knockout (accuracy drops to near-chance, 12–19%), sufficiency via Fisher activation patching (9D transfers 83–100% of digit information), and specificity via random controls (0% effect at all dimensions). This constitutes a complete "evidentiary triangle" (§4.2).

3. **Progressive rotation.** We discover that models progressively rotate the Fourier subspace across layers to align with the unembedding matrix W_U, with alignment increasing from ∼30% at computation layers to 100% at readout layers (§4.3).
4. **Architecture-specific CRT components.** Different models emphasize different frequencies of the Chinese Remainder Theorem (CRT) decomposition $\mathbb{Z}/10\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$: Gemma uses k=5 (parity), Phi-3 uses k=2 (mod-5), and LLaMA uses a balanced ordinal encoding (§4.4).
5. **Fourier phase rotation steering.** We demonstrate that rotating activations within the Fourier subspace can steer digit predictions with 70–88% exact shift accuracy, and identify the encoding-readout gap as the bottleneck overcome by unembedding-informed steering (§4.5).

2 Background and Related Work

Mechanistic interpretability of arithmetic. Nanda et al. [2023] showed that small transformers trained on modular addition learn Fourier representations, using pairs of sine and cosine features to implement the "clock algorithm." Zhong et al. [2024] extended this to general group operations. Our work differs in studying *pretrained* language models performing natural-language arithmetic, where the Fourier structure emerges without explicit modular arithmetic training.

Activation patching and causal methods. Activation patching [Vig et al., 2020, Meng et al., 2022, Geiger et al., 2024] replaces activations from one input with those from another to measure causal effects. We use both full activation patching (for layer scanning) and subspace-restricted patching (Fisher and Fourier directions) to establish causal sufficiency.

Fourier analysis on finite groups. The discrete Fourier transform on $\mathbb{Z}/N\mathbb{Z}$ decomposes functions into N frequency components. For N=10, the irreducible representations give frequencies $k \in \{0, 1, 2, 3, 4, 5\}$ with k=0 being the mean (discarded after centering), k=$1, \ldots, 4$ each contributing 2 dimensions (cosine and sine), and k=5 (Nyquist) contributing 1 dimension, totaling $1 + 2{\times}4 + 1 = 10$ dimensions with the mean, or 9 dimensions after centering.

Chinese Remainder Theorem. Since $10 = 2 \times 5$ with $\gcd(2, 5) = 1$, the CRT gives $\mathbb{Z}/10\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$. The frequency k=5 corresponds to $\mathbb{Z}/2\mathbb{Z}$ (parity: $(-1)^d$), while k=2 corresponds to the generator of $\mathbb{Z}/5\mathbb{Z}$ (period-5 structure). This decomposition is relevant because models could, in principle, compute ones-digit addition by separately computing mod-2 and mod-5 components and combining via CRT.

3 Methods

3.1 Models and Setup

We study three pretrained models: **Gemma 2B** (google/gemma-2-2b, 26 layers, d_{model}=2304), **Phi-3 Mini** (microsoft/Phi-3-mini-4k-instruct, 32 layers, d_{model}=3072), and **LLaMA 3.2-3B** (meta-llama/Llama-3.2-3B, 28 layers, d_{model}=3072). All experiments use TransformerLens [Nanda and Bloom, 2022] for hook-based interventions.

Problem generation. We generate two-operand addition problems in "`a + b = c`" format. Gemma and Phi-3 use *teacher-forced* multi-digit problems where we provide the tens digit and measure ones-digit prediction. LLaMA uses *direct-answer* single-digit problems ($a + b < 10$) since LLaMA tokenizes all integers 0–198 as single tokens, enabling clean single-step prediction. All problems are balanced to ∼100 examples per ones digit (0–9).

Layer identification. We perform full-layer activation patching scans to identify *computation layers* (where arithmetic information first appears) and *readout layers* (where the model commits to an answer). Table 1 summarizes.

Table 1: Key layers identified by full-patch transfer scan. *Computation* = first layer with >50% transfer; *Readout* = first layer with 100% transfer.

Model	**Total Layers**	**Computation**	**Readout**	d_{model}	d_{mlp}
Gemma 2B	26	L18 (65%)	L21 (100%)	2304	9216
Phi-3 Mini	32	L21 (76%)	L28 (100%)	3072	8192
LLaMA 3.2-3B	28	L16 (99%)	L22 (100%)	3072	8192

3.2 Fourier Basis Extraction

Given a set of correctly-solved problems, we collect the residual stream activation $\mathbf{h}_\ell \in \mathbb{R}^{d_{\text{model}}}$ at layer ℓ at the position where the ones digit is predicted. We compute per-digit means $\boldsymbol{\mu}_d = \mathbb{E}[\mathbf{h}_\ell \mid \text{ones digit} = d]$ for $d \in \{0, \ldots, 9\}$, forming a $10 \times d_{\text{model}}$ matrix M.

We extract the 9D Fourier subspace via SVD of the row-centered M: $\bar{M} = U\Sigma V^\top$, where $U \in \mathbb{R}^{10\times 9}$ contains the digit patterns and $V \in \mathbb{R}^{d_{\text{model}}\times 9}$ contains the subspace directions. To label the frequency content of each SVD direction, we compute the DFT of each column of U and assign the dominant frequency k^*.

Hybrid SVD+DFT basis. For phase rotation experiments (§4.5), we use a hybrid method: SVD provides the orthogonal 9D subspace, then DFT within the SVD coordinate system identifies cosine/sine axes for each frequency. This guarantees both orthonormality (from SVD) and correct frequency labeling (from DFT).

Frequency purity. For each SVD direction with DFT spectrum $\{|c_k|^2\}_{k=1}^5$, purity is $\max_k |c_k|^2 / \sum_k |c_k|^2$. A purity of 100% means the direction is a pure sinusoid at one frequency.

3.3 Causal Methods

Full activation patching. For each pair of problems $(p_{\text{clean}}, p_{\text{corrupt}})$ with different ones digits, we run p_{clean}, cache all activations, then run p_{corrupt} while replacing the residual stream at layer ℓ with the cached clean activation. *Transfer* = fraction of pairs where the patched model outputs the clean answer.

Subspace-restricted patching. Given an orthonormal basis $B \in \mathbb{R}^{d_{\text{model}}\times k}$ for a k-dimensional subspace, we patch only the projection: $\mathbf{h}_{\text{patched}} = \mathbf{h}_{\text{corrupt}} + BB^\top(\mathbf{h}_{\text{clean}} - \mathbf{h}_{\text{corrupt}})$. We use this for Fisher, contrastive Fisher, and unembedding-aligned subspaces.

Fisher subspace. We compute the Fisher information matrix $F = \mathbb{E}[\nabla_{\mathbf{h}} \log p(y|\mathbf{h}) \nabla_{\mathbf{h}} \log p(y|\mathbf{h})^\top]$ at layer ℓ and take its top eigenvectors. *Contrastive Fisher* averages the Fisher matrix within each digit class and takes the top eigenvectors of the between-class Fisher. All Fisher computations use CPU to avoid MPS gradient artifacts (Appendix A).

Fourier knockout. We zero out the 9D Fourier projection at one or more layers: $\mathbf{h}_{\text{ablated}} = \mathbf{h} - VV^\top\mathbf{h}$, where V spans the Fourier subspace. We also perform frequency-specific knockout (ablating only the 1–3 dimensions assigned to a single frequency k) and random controls (ablating matched-dimension random subspaces).

3.4 Fourier Phase Rotation

To test whether the Fourier encoding supports controllable digit manipulation, we *rotate* the Fourier component of activations to shift the encoded digit by j positions (mod 10). For each frequency k with cosine/sine axes $(\mathbf{e}_k^c, \mathbf{e}_k^s)$, we apply:

$$R_k(j) = \begin{pmatrix} \cos(2\pi kj/10) & -\sin(2\pi kj/10) \\ \sin(2\pi kj/10) & \cos(2\pi kj/10) \end{pmatrix} \tag{1}$$

applied to the (c_k, s_k) coordinates of the activation projected onto the Fourier subspace. The rotation is applied coherently across all frequencies simultaneously. We test three steering variants:

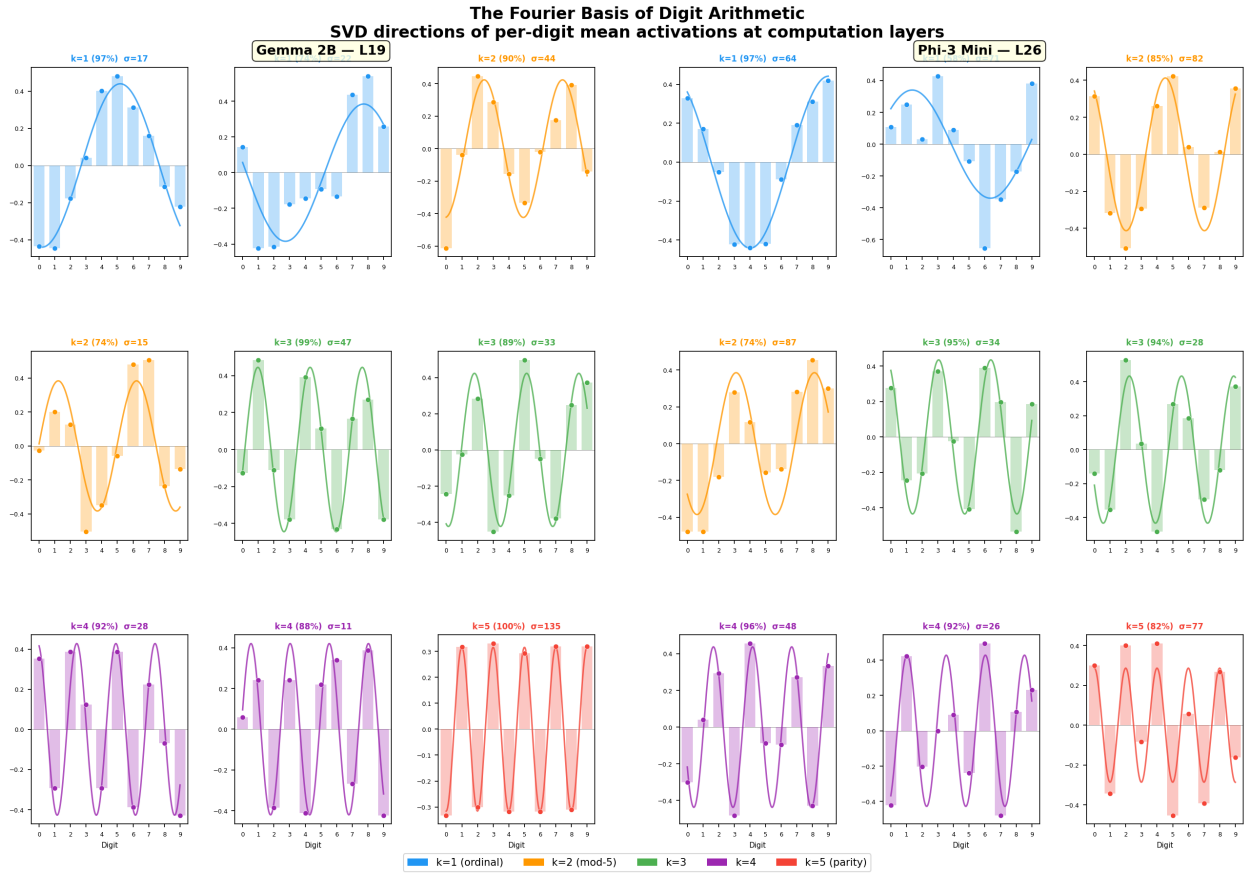

Figure 1: **SVD directions of per-digit mean activations are discrete Fourier modes.** Each panel shows one SVD direction's loadings on digits 0–9 (bars) overlaid with the best-fit DFT sinusoid (curve). Frequency assignment and purity shown above each panel. Left: Gemma 2B at L19. Right: Phi-3 Mini at L26. All directions achieve >70% purity, with most >85%.

- **Coherent**: Apply the rotation delta $\Delta = R(\mathbf{h}_{\text{fourier}}) - \mathbf{h}_{\text{fourier}}$ scaled by factor N.
- **W_U-projected**: Project Δ onto the W_U column space before adding.
- **W_U-steer**: Use $\mathbf{w}_{\text{target}} - \mathbf{w}_{\text{orig}}$ from W_U digit columns, projected onto the Fourier subspace and normalized to $\|\Delta\|$, scaled by N.

4 Results

4.1 The 9D Fourier Subspace

At computation layers, the SVD of per-digit mean activations reveals a *perfect Fourier basis* of $\mathbb{Z}/10\mathbb{Z}$: each of the 9 SVD directions is dominated by a single DFT frequency, and every frequency $k \in \{1, \ldots, 5\}$ receives exactly the expected number of degrees of freedom (2 for k=1, ..., 4; 1 for k=5). Figure 1 shows the SVD direction digit loadings overlaid with their best-fit sinusoids.

Table 2 summarizes frequency purities across models. All three models show frequency purities of 72–99% at their computation layers, indicating that the learned digit representations are well-described by discrete Fourier modes.

CP tensor decomposition confirms trig identity structure. We decompose the per-digit outer product tensor $T_{d,i,j} = \mathbb{E}[h_i \cdot h_j \mid \text{digit}{=}d]$ projected into the Fourier subspace. The σ^2-weighted trigonometric

Table 2: Frequency purity of SVD directions at computation layers.

Model	Layer	Mean Purity	Dominant Dir	σ_1/σ_9	Perfect Basis?
Gemma 2B	L19	89.2%	k=5, σ=135	7.4×	✓
Phi-3 Mini	L26	85.9%	k=2, σ=87	3.1×	✓
LLaMA 3.2-3B	L20	67.1%	k=5, σ=8.3	2.5×	✓

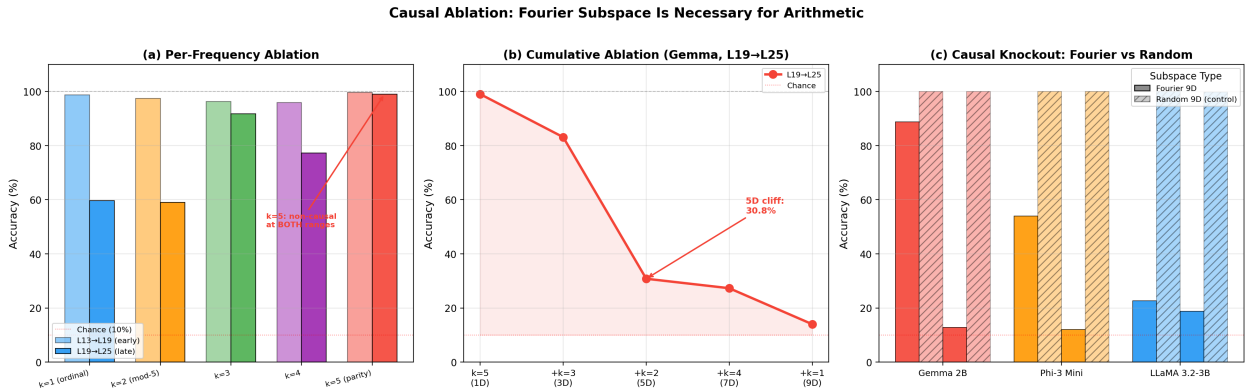

Figure 2: **Causal ablation results.** (a) Per-frequency multi-layer ablation on Gemma 2B: k=5 (parity) is non-causal at both early and late layer ranges, while k=1 and k=2 each cause ∼40% damage at late layers. (b) Cumulative ablation ordered by σ^2 at L19–L25: a sharp cliff at 5D. (c) Cross-model Fourier knockout: multi-layer ablation drops all models to ≤19%, while random 9D ablation has zero effect.

identity score is 0.964 for Gemma, confirming that the conditional expectation tensor satisfies $\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$—the hallmark of Fourier-domain angle addition.

4.2 Causal Necessity and Sufficiency

Necessity: Fourier knockout. Zeroing the 9D Fourier subspace across all layers from computation to readout reduces accuracy to near-chance for all three models (Table 3). Critically, matched-dimension random subspace ablation has *zero* effect, establishing perfect specificity. Single-layer ablation causes partial damage (11–77%), revealing a distributed pipeline where Fourier information is actively maintained across multiple layers.

Sufficiency: Fisher activation patching. At readout layers, Fisher subspace patching transfers nearly all digit information (Table 4). Standard Fisher at 10D transfers 85% (Gemma, Phi-3) and 100% (LLaMA); contrastive Fisher at 9D transfers 83–100%. The alignment between standard Fisher top-20D and contrastive Fisher 9D subspaces exceeds 0.97 in principal cosine, confirming they identify the same subspace.

The k=5 paradox. Despite being the dominant SVD direction at Gemma's computation layer (σ=135, 71% of subspace variance), k=5 (parity) is *causally inert*: multi-layer ablation of k=5 across L13–L25 causes only 0.4–1.0% accuracy damage. In contrast, k=1 and k=2 each cause ∼40% damage when ablated at L19–L25. This establishes that k=5 is an **epiphenomenal encoding**—the model encodes parity strongly but does not use it for ones-digit computation. The causally necessary frequencies are k=1 (ordinal) and k=2 (mod-5).

4.3 Progressive Rotation into Readout Position

We track how the digit subspace aligns with the unembedding matrix W_U across layers using unembed-aligned patching. The results reveal **progressive rotation**: at computation layers, the Fourier subspace captures digit information that is largely *orthogonal* to W_U directions, but by readout layers, the same information has rotated to align perfectly with W_U.

Table 3: Fourier knockout results. Multi-layer ablation covers all layers from computation to readout. Chance $\approx 10\%$.

Model	**Single Fourier 9D**	**Single Random 9D**	**Multi Fourier 9D**	**Multi Random 9D**
Gemma 2B	88.7% (−11.3%)	100% (0%)	12.8% (−87.2%)	100% (0%)
Phi-3 Mini	54.0% (−46.0%)	100% (0%)	12.1% (−87.9%)	100% (0%)
LLaMA 3.2-3B	22.7% (−77.3%)	100% (0%)	18.8% (−81.2%)	99.8% (0%)

Table 4: Fisher activation patching transfer at readout layers (% of full-patch transfer).

	Standard Fisher					**Contrastive Fisher**		
Model (Layer)	2D	5D	10D	20D	50D	2D	5D	9D
Gemma L21	4.1	20.7	85.2	96.3	99.3	5.9	29.3	82.6
Phi-3 L28	2.2	17.8	85.2	99.6	100	5.9	34.1	93.7
LLaMA L27	3.2	35.8	100	100	100	8.7	56.4	100

For LLaMA, unembed-aligned 9D patching transfers: L20 (29.8%) → L21 (56.9%) → L24 (94.5%) → L27 (100%). The complementary orthogonal patching decreases correspondingly: 86.7% → 13.8% → 0.0% → 0.0%. This progressive alignment is consistent across all three models.

Dissociation between Fisher and unembed alignment. At intermediate layers, Fisher patching captures substantial digit information (e.g., Gemma L21: Fisher 10D = 85%) while unembed patching captures very little (Gemma L21: unembed 9D = 30%). This dissociation means the digit encoding is gradient-visible but not yet output-aligned—Fisher and unembedding rotation are *separate processes* that converge only at output layers.

4.4 Architecture-Specific CRT Components

The Fourier energy spectrum varies dramatically across layers and models (Figure 4):

- **Gemma 2B**: k=5 (parity) dominates at L18–L19 (64–71% of energy), then drops sharply at L20+ as k=1 takes over. The model computes parity *first*, then transforms to ordinal encoding.
- **Phi-3 Mini**: k=2 (mod-5) surges at L24–L26 (40%), with k=1 ordinal dominant at earlier layers. Phi-3 computes the mod-5 CRT component during its main computation phase.
- **LLaMA 3.2-3B**: k=1 + k=2 dominate at L16–L18 (74–78% combined), with k=3 and k=4 *suppressed* to ∼1%. LLaMA uses balanced ordinal+mod-5 encoding from the start.

All models converge to approximately ordinal encoding (k=1 > k=2 > ⋯ > k=5) at readout layers, consistent with the unembedding matrix being organized by digit magnitude.

Per-neuron frequency tuning. At computation layers, a substantial fraction of MLP neurons are tuned to individual Fourier frequencies (Figure 5). In Gemma L19, 809/9216 neurons (8.8%) have >80% frequency purity, with k=1 and k=5 as the dominant tunings.

Energy explosion. Total Fourier energy in the digit subspace increases by 2–3 orders of magnitude at computation layers (Figure 6), indicating active amplification of digit-discriminative signals rather than passive propagation.

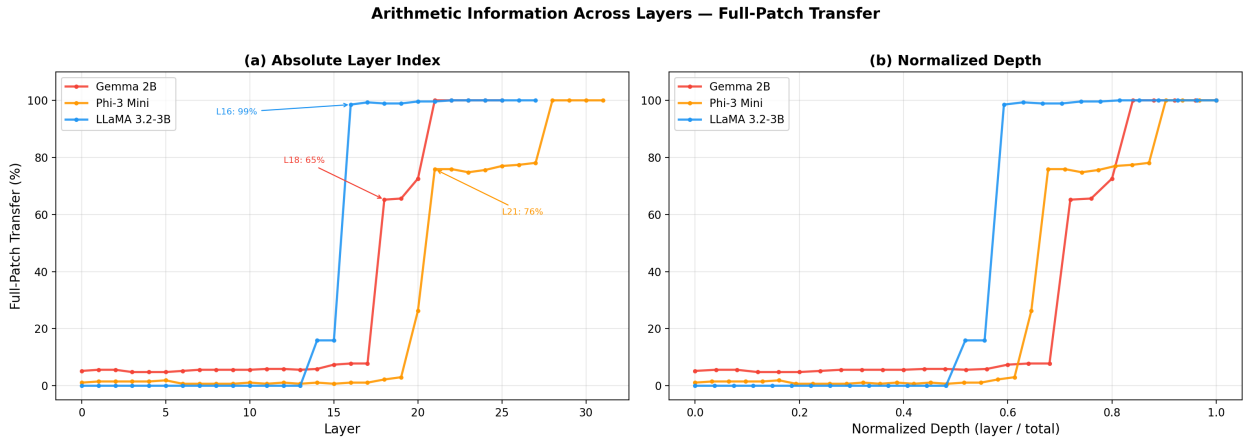

Figure 3: **Arithmetic information emerges at sharp phase transitions.** (a) Absolute layer indices show model-specific transition points: L18 for Gemma, L16 for LLaMA, L21 for Phi-3. (b) Normalized depth reveals transitions occur at similar relative positions (∼0.55–0.70 depth).

Table 5: Progressive rotation: unembed-aligned 9D patching transfer across layers (LLaMA 3.2-3B).

	L20 (comp)	**L21**	**L24**	**L27 (readout)**
Unembed 9D transfer	29.8%	56.9%	94.5%	100.0%
Orthogonal transfer	86.7%	13.8%	0.0%	0.0%

4.5 Fourier Phase Rotation Steering

We test whether rotating activations within the Fourier subspace can steer digit predictions. Given a problem with correct answer digit d, we apply Fourier rotation to shift the encoding to target digit $(d + j) \bmod 10$ for each shift $j \in \{1, \ldots, 9\}$.

Baseline coherent rotation. At computation layers with the hybrid SVD+DFT basis, coherent rotation achieves above-chance steering for all models, with strong backward-shift asymmetry: Gemma L19 averages 28% exact (range: j=1: 27%, j=9: 62%), Phi-3 L26 averages 28% (range: j=1: 49%, j=9: 56%), and LLaMA L20 averages 69% (range: j=1: 82%, j=9: 94%).

The encoding-readout gap. The Fourier-Unembed overlap (fraction of Fourier subspace variance captured by W_U digit directions) is only 8–11% across models at computation layers. This means rotating in Fourier space does not directly map to the readout directions W_U uses. This gap is the primary bottleneck for steering performance.

W_U-informed steering bridges the gap. The W_U-steer method (using $\mathbf{w}_{\text{target}} - \mathbf{w}_{\text{orig}}$ projected onto the Fourier subspace) dramatically improves steering:

W_U-steer is transformative for Gemma (16% → 70%) and Phi-3 (14% → 84%), where plain rotation was bottlenecked by the encoding-readout gap. For LLaMA, where Fourier structure is already highly pure (93–99%), simple coherent scaling is nearly as effective. W_U-steer also eliminates the forward/backward asymmetry: Phi-3 at ×5 gives j=1: 82%, j=9: 96% (was 20%/29% at ×1 coherent).

Readout layers are immune. At readout layers (Gemma L25, LLaMA L27), all steering methods produce ≤10% exact shift (chance level). The Fourier-Unembed overlap drops to 0.2% at LLaMA L27, confirming that by readout, the Fourier subspace has been fully absorbed into the unembedding-aligned representation and can no longer be independently manipulated.

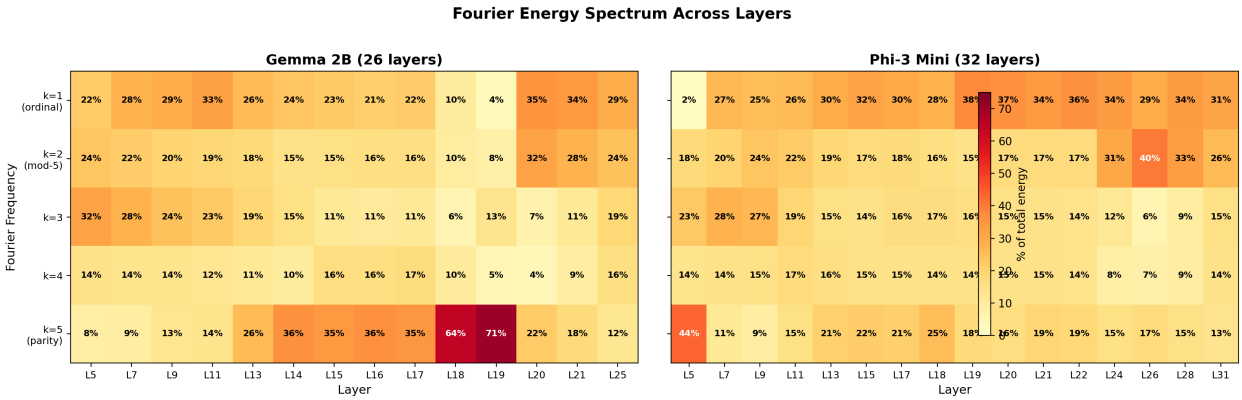

Figure 4: **Fourier energy spectrum across layers.** Gemma (left) shows a dramatic k=5 (parity) spike at L18–L19, while Phi-3 (right) shows a k=2 (mod-5) surge at L24–L26. Both converge to ordinal (k=1 dominant) at readout.

Table 6: Steering exact digit shift rates (%) with different methods. Best scale factor N shown.

Method	**Gemma L19**	**Phi-3 L26**	**LLaMA L20**
Coherent $\times 1$	16.0	14.3	69.2
Coherent $\times 3$	—	—	86.4
W_U-proj $\times 10$	—	—	88.1
W_U-steer $\times 5$	**69.7**	**84.2** ($\times 10$)	82.2 ($\times 3$)
Clean logit-lens acc.	52.7%	100%	100%

5 Discussion

Fourier structure as a universal arithmetic primitive. Our finding that three independently trained architectures all converge on a Fourier basis of $\mathbb{Z}/10\mathbb{Z}$ for digit encoding suggests this is not an architectural accident but a natural computational strategy. The Fourier basis is mathematically optimal for representing cyclic structure: it diagonalizes the shift operator, making addition a simple phase rotation. That pretrained LLMs discover this representation without explicit modular arithmetic training mirrors findings in trained "grokking" models [Nanda et al., 2023], suggesting a deep connection between learned and theoretically optimal representations.

The k=5 paradox and epiphenomenal encoding. The striking finding that k=5 (parity) dominates Gemma's computation layer yet is causally unnecessary challenges a naive interpretation of representational analyses. The model encodes parity information strongly—perhaps because it is useful for other tasks or emerges naturally from the training distribution—but does not use it for ones-digit addition. This highlights the importance of complementing observational analyses (SVD, Fourier decomposition) with causal interventions (knockout, patching). Without the multi-layer frequency ablation experiment, we would have incorrectly concluded that parity is a central computational feature.

Progressive rotation as a computational mechanism. The progressive rotation of digit information from an arbitrary encoding subspace into the W_U-aligned readout subspace represents a novel mechanistic finding. This suggests that intermediate layers serve as a "rotation pipeline" that transforms representations from a format optimized for computation (the Fourier basis, which diagonalizes the shift operator) into a format optimized for readout (the unembedding basis, which maps to logits). The dissociation between Fisher visibility and W_U alignment at intermediate layers (e.g., Gemma L21: Fisher 10D captures 85% but W_U 9D captures only 30%) provides direct evidence that these are distinct representational processes.

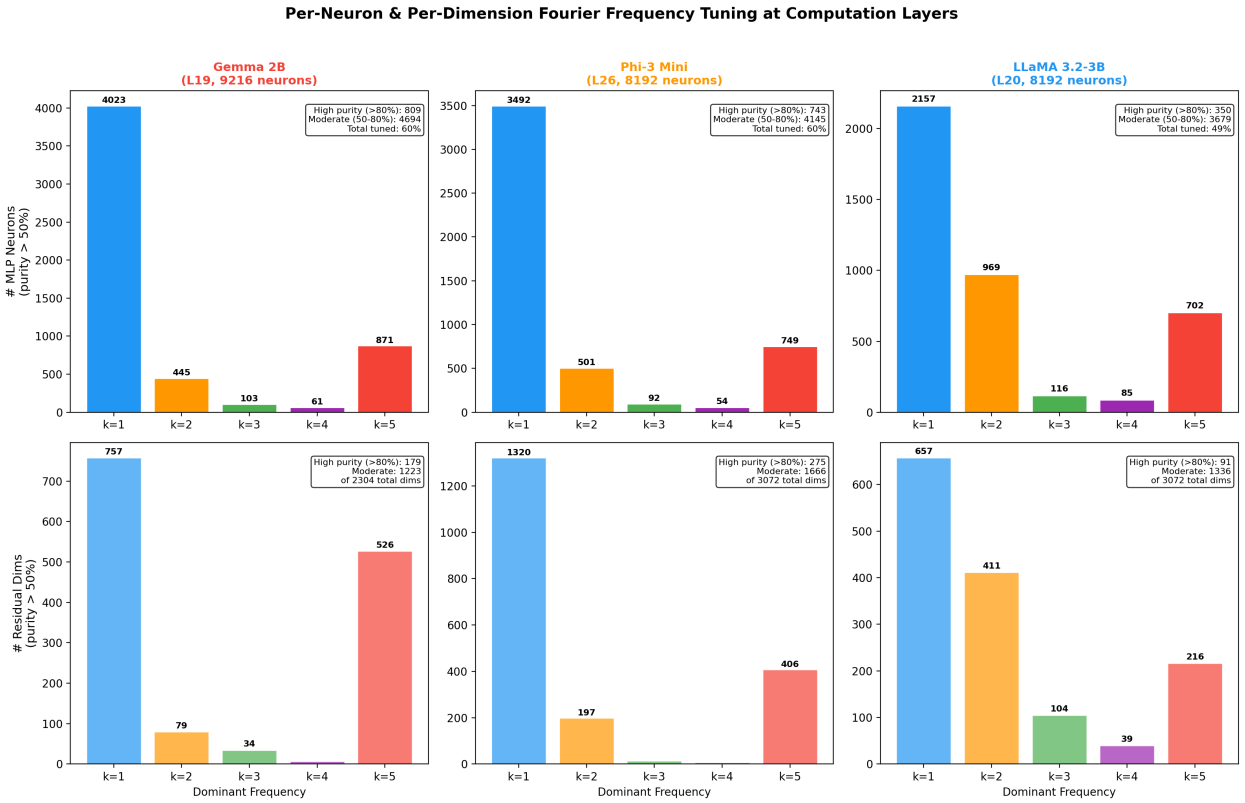

Figure 5: **Per-neuron and per-dimension Fourier frequency tuning** at computation layers. Top: MLP neurons with >50% purity. Bottom: residual stream dimensions. All models show k=1 (ordinal) as the most common tuning, with model-specific secondary peaks.

Architecture-specific CRT strategies. The fact that Gemma emphasizes parity (k=5, corresponding to $\mathbb{Z}/2\mathbb{Z}$), Phi-3 emphasizes mod-5 (k=2, corresponding to $\mathbb{Z}/5\mathbb{Z}$), and LLaMA uses a balanced approach suggests that the CRT decomposition $\mathbb{Z}/10\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$ is a useful framework for understanding arithmetic circuits, even though no model implements a clean sequential CRT algorithm. The architectural differences likely stem from differences in training data, model size, and layer count affecting which computational strategies are favored during training.

Steering and the encoding-readout gap. The success of W_U-informed steering (70–88% exact shift) demonstrates genuine mechanistic control over arithmetic computation. The key insight is that the Fourier subspace and the unembedding readout are related but not identical: the Fourier-Unembed overlap is only 8–11%. The W_U-steer method bridges this gap by directly targeting the readout direction while constraining the intervention to the Fourier subspace. The residual 12–30% failure rate is attributable to LayerNorm nonlinearity, within-class variance in the Fourier projection, and the inherent limitation of single-layer interventions in a distributed pipeline.

Limitations.

- Our analysis is restricted to ones-digit addition with operands < 100 (or < 10 for LLaMA). Multi-digit carry propagation circuits remain unexplored.
- The Fourier basis is extracted from per-digit *means*, discarding within-digit variance that may carry information about operand identity, carry status, or tens digit.
- Fisher patching on Apple Silicon MPS produces incorrect results (Appendix A); all Fisher results use CPU. This limits scalability to larger models.

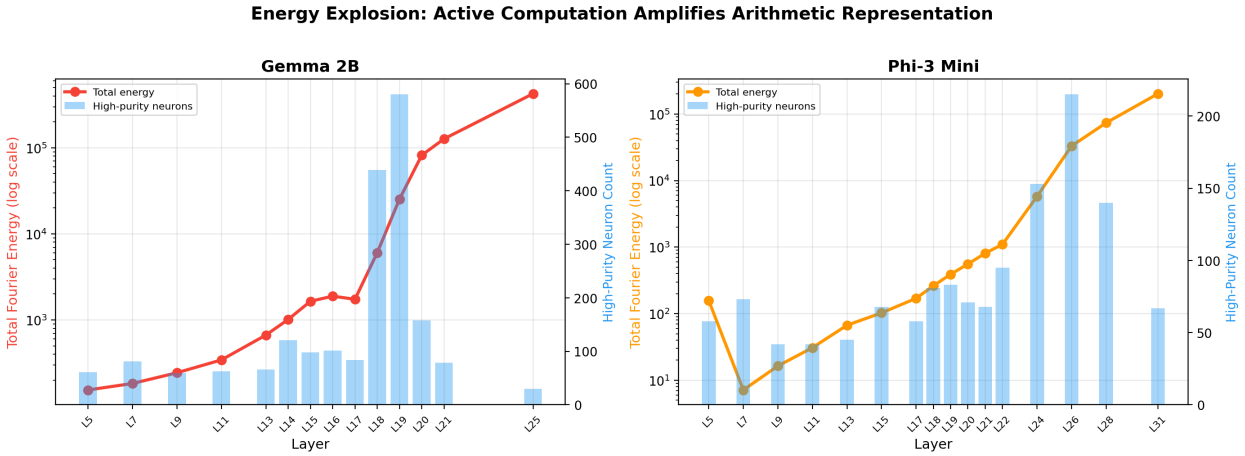

Figure 6: **Energy explosion** at computation layers. Total Fourier energy (log scale) and high-purity neuron count both spike at computation onset.

- We study only three models from different families. Within-family scaling laws (e.g., Gemma 2B vs 7B vs 27B) would strengthen generalization claims.

Future work. Natural extensions include: (1) analyzing carry propagation circuits using the same Fourier framework; (2) testing whether the Fourier basis generalizes to multiplication and other modular operations; (3) studying how the Fourier subspace emerges during training (grokking dynamics in full-scale models); and (4) using the identified circuit for targeted model editing (e.g., correcting arithmetic errors by intervening on the Fourier subspace).

6 Conclusion

We have shown that pretrained language models encode ones-digit identity in a 9-dimensional Fourier subspace of $\mathbb{Z}/10\mathbb{Z}$, established its causal necessity and sufficiency via a complete ablation-patching-specificity evidence triangle, discovered the progressive rotation mechanism by which this subspace aligns with the unembedding matrix, characterized architecture-specific CRT component emphasis, and demonstrated controllable steering of digit predictions via Fourier phase rotation. These findings provide a detailed mechanistic account of how language models perform integer addition, grounded in the representation theory of cyclic groups.

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A MPS Gradient Artifact

During development, we discovered that Fisher information matrix computation on Apple Silicon MPS backend produces qualitatively different (and incorrect) eigenvectors compared to CPU computation. For Gemma 2B at L22, MPS Fisher patching at 50D yielded 78.8% transfer vs 20.1% on CPU—a 4× inflation. The MPS effective dimensionality was also inflated (16.08 vs 7.32 on CPU). We attribute this to numerical precision differences in gradient accumulation on the MPS backend. All Fisher results reported in this paper use CPU computation.

B Component Attribution

We performed Fourier-projected Direct Logit Attribution (DLA) and single-component causal patching across all attention heads and MLP layers for each model. Table 7 summarizes the circuit architecture.

Table 7: Component attribution summary. DLA = Fourier-projected direct logit attribution (% of total Fourier energy). Damage = accuracy drop from single-component mean ablation.

Model	**Circuit Type**	**Top DLA Component**	**Causally Necessary**	**Damage**
Gemma 2B	Pure MLP	MLP L19 (29.2%)	MLP L19	0.7%
Phi-3 Mini	Mixed	MLP L25 (20.4%)	L21H30	27.9%
			MLP L25	1.8%
LLaMA 3.2-3B	Pure MLP	MLP L20 (23.8%)	MLP L16	14.1%

Gemma and LLaMA circuits are MLP-dominated, with no attention head showing measurable causal impact. Phi-3 is the only model with a causally necessary attention head (L21H30, 27.9% damage), located well before the computation layer (L26), suggesting an early-stage information routing role. Despite having the highest DLA, MLP L20 in LLaMA is individually dispensable; MLP L16 (the computation bottleneck) is the only causally necessary component.

C Head Attribution Frequency Breakdown

The causally necessary components show characteristic frequency tunings that align with the model-specific CRT emphasis:

- **Gemma MLP L19**: k=5 (46%) – parity-dominant, matching the Fourier layer sweep finding of k=5 dominance at L19.
- **Phi-3 L21H30**: Broadly tuned (k=3: 22%, k=2: 21%, k=1: 19%) – consistent with a general "arithmetic attention" role.
- **LLaMA MLP L16**: k=1 (40%) + k=2 (33%) – ordinal + mod-5, matching the computation onset Fourier spectrum.

D Fourier Phase Rotation Details

Forward/backward asymmetry. Coherent rotation shows a systematic asymmetry: backward shifts (j=7, 8, 9) achieve higher exact rates than forward shifts (j=1, 2, 3). Diagnosis reveals this stems from non-uniform Fourier-Unembed overlap across digits: digits 3, 5, 6 have 13–15% overlap (basins of attraction), while digits 0, 1 have only 1–2% overlap. Backward shifts for common answer digits target high-overlap digits, while forward shifts target low-overlap digits. The W_U-steer method largely eliminates this asymmetry by directly targeting W_U readout directions.

Readout layer immunity. At readout layers (Gemma L25, LLaMA L27), all steering methods produce ≤10% exact shift rate. The Fourier-Unembed overlap drops from ∼10% at computation layers to 0.2% at LLaMA L27, confirming that by readout, the Fourier subspace has been fully absorbed into the W_U-aligned representation and can no longer be independently manipulated.

Elliptical correction. The SVD singular value ratios within each frequency pair (cos/sin) range from 1.05× (LLaMA, near-circular) to 3.04× (Gemma, highly elliptical). Elliptical correction—scaling each axis by its singular value before rotation—lifted exact rates from ∼0% to 8–15% for all models, confirming that the encoding geometry is genuinely elliptical rather than circular.

E Cumulative Ablation Data

Table 8: Cumulative multi-layer frequency ablation on Gemma 2B. Frequencies added in order of decreasing σ^2 at L19. L13–L19 = early computation; L19–L25 = late computation+readout.

Frequencies Ablated	**Total Dims**	**Acc (L13–L19)**	**Acc (L19–L25)**
Baseline	0	100.0%	100.0%
k=5	1	99.6%	99.0%
+k=3	3	—	83.1%
+k=2	5	—	30.8%
+k=4	7	—	27.3%
+k=1 (all 9D)	9	52.8%	14.0%
Random 9D (all layers)	9	100.0%	100.0%